

Thailand Road Accident Analysis (2019–2022)

Predicting Accident Severity from Environmental and Contextual Factors

CPE393, Introduction to Data Science with Python
King Mongkut's University of Technology Thonburi (KMUTT)
Group BRQ

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GitHub repository: <https://github.com/RomainBardet/FinalProjetDataScience>

1. Problem Definition and Goals

Road accidents are a serious public safety problem in Thailand, and they cause important human and economic losses every year. National dashboards usually only show raw accident counts, and do not really try to predict which conditions lead to the most severe outcomes. This is the gap we wanted to study: we analyzed 81,735 road accident records collected across Thailand between 2019 and 2022, and we built a model that predicts the severity of an accident using only the environmental and contextual information available at the time it happens.

We built accident severity as a three-class label from the fatality and injury counts of the dataset: “Fatal” if at least one person died, “Injury” if someone was injured but nobody died, and “Property Damage Only (PDO)” if neither happened. This makes the project a multi-class classification problem, which corresponds to the project types recommended in the course guidelines.

1.1 Target question

Can we predict the severity of a road accident in Thailand using only the information available directly at the scene: vehicle type, weather, road and slope description, time of day, day of week, province, and the agency in charge, without using the outcome variables themselves?

1.2 Why this matters

- Identifying which combinations of conditions are linked to fatal outcomes can help prioritize road safety interventions (signage, lighting, enforcement) where they are needed the most.
- Understanding which vehicle types and road configurations are over-represented in severe accidents can guide vehicle safety regulation and driver education campaigns.
- A transparent model, evaluated honestly, can support policy discussions without exaggerating what the data can actually predict, a Responsible AI point we discuss again in Section 9.

Expected outcome: a comparison of three classification models, an honest evaluation of their ability (and their limits) to detect fatal accidents, and a discussion of what this means for road safety policy in Thailand.

2. Dataset Description

The dataset, “Thailand Road Accident (2019–2022)”, is publicly available on Kaggle. It originally comes from raw accident records published by the Office of the Permanent Secretary, Ministry of Transport, Thailand, and it is also used in a public road-safety dashboard, which supports its credibility as a source.

Property	Value
Source	Kaggle (originally Thailand Ministry of Transport)
Format	CSV
Size	81,735 rows × 18 columns

License / usage	Public Kaggle dataset, used here for educational purposes
Target variable	accident_severity (engineered: Fatal / Injury / PDO)
Time span	2019–2022 (incident_datetime)

2.1 Variables

The raw dataset contains the following columns:

Column	Description
acc_code	Unique accident identifier
incident_datetime	Date and time the accident occurred
report_datetime	Date and time the accident was officially reported
province_th / province_en	Province name, in Thai and English
agency	Government agency responsible for the road
route	Road segment / route description
vehicle_type	Type of vehicle primarily involved
presumed_cause	Officially presumed cause of the accident
accident_type	Nature of the collision (exemp, rear-end, rollover)
number_of_vehicles_involved	Count of vehicles involved
number_of_fatalities	Count of deaths (used to build the target)
number_of_injuries	Count of injuries (used to build the target)
weather_condition	Weather at the time of the accident
latitude / longitude	GPS coordinates of the accident site
road_description	Road geometry (exemp, straight, curved)
slope_description	Slope condition of the road

2.2 Why this dataset is interesting

We chose this dataset because it is large enough to train real ML models (more than 80,000 records), and because it covers a concrete real-world problem that felt relevant to us during our exchange in Thailand. It also contains both categorical and numeric variables, which allowed us to practice the full data science pipeline taught in this course.

2.3 Known dataset limitations (identified before modeling)

Before making any cleaning decisions, we performed an initial audit of the raw data and found the following issues:

Issue	Rows affected	Severity
report_datetime delayed more than 30 days vs. incident_datetime (median delay: 48.7 days)	≈48,400 (59%)	Critical
report_datetime earlier than incident_datetime (logical inconsistency)	1	Critical
Missing GPS coordinates	359 (0.44%)	Critical
GPS coordinates outside Thailand's territory	3	Critical
number_of_vehicles_involved = 0 (physically inconsistent)	109	Significant
presumed_cause dominated by "speeding" (possible reporting bias)	≈60,000 (74%)	Significant
Catch-all "other" category in several columns	varies	Minor

We used these findings to decide what to clean and what to keep. The main conclusion was that report_datetime cannot be used for any time-based analysis, so we used incident_datetime instead.

3. Preprocessing and Data Cleaning

3.1 Duplicate records

We checked for fully duplicated rows and for duplicated acc_code identifiers, and found none: 0 duplicated rows and 0 duplicated accident codes among all 81,735 records. No deduplication step was therefore necessary.

3.2 Incorrect data types

When importing the data, incident_datetime and report_datetime are read as plain text instead of proper datetime objects, which would make any date computation (for example, computing the reporting delay or extracting the hour of the accident) unreliable or impossible. We converted both columns to pandas datetime type using pd.to_datetime() before any further analysis. All other columns already had an appropriate type on import (integers for counts, floats for coordinates, strings for categories), so no further conversion was necessary.

3.3 Outliers

We examined number_of_vehicles_involved, number_of_fatalities, and number_of_injuries for outliers using the IQR method. A strict IQR rule would flag a large proportion of rows as outliers on all three columns: for example, any accident with one or more fatalities technically falls

outside the IQR bound of 0, since the median and most accidents have zero fatalities. We considered that removing these rows would not be appropriate: a multi-vehicle accident with several fatalities is a real and valid case, and arguably the most policy-relevant type of accident in the dataset, not a data entry error. We therefore did not remove statistical outliers from these three columns, and instead applied the targeted cleaning rule described in Section 2.3: removing the 109 rows with `number_of_vehicles_involved = 0`, which is the only value in these columns that is logically impossible rather than simply rare.

3.4 Removing physically inconsistent rows

- We removed 109 rows with `number_of_vehicles_involved = 0`: a recorded road accident cannot logically involve zero vehicles, so we considered these rows as data entry errors rather than a valid category.
- We also removed 3 rows with GPS coordinates located outside Thailand's approximate bounding box (latitude 5°–21°N, longitude 97°–106°E), since they are very likely coordinate entry errors and cannot be used for geospatial analysis.

3.5 Handling missing values

359 rows (0.44%) have missing GPS coordinates. Since this represents a small fraction of the data and does not affect non-spatial columns, we kept these rows for all non-geospatial analysis and modeling, and excluded them only from the geospatial scatter plot and from any latitude/longitude-based feature.

3.6 Correcting inconsistent formats

All categorical text columns were stripped of leading and trailing whitespace and converted to lowercase, to avoid treating "Clear" and "clear " as different categories during one-hot encoding.

3.7 Choosing the reliable timestamp

Since 59% of accidents were reported more than 30 days after they occurred (and one report even precedes the incident itself), `report_datetime` cannot be trusted to represent when accidents actually happen. All time-based features (year, month, hour, day of week, weekend flag) were therefore derived from `incident_datetime` only.

3.8 Engineering the target variable

`accident_severity` was built directly from `number_of_fatalities` and `number_of_injuries`, as described in Section 1. After cleaning, the class distribution is reasonably balanced for a real-world severity label: PDO 44.4%, Injury 44.5%, and Fatal 12.0%. This means the minority class (Fatal) is present in sufficient volume to be learnable, even though it remains the hardest and most important class to detect correctly.

3.9 Dropping non-predictive or leakage-prone columns

Before modeling, we removed `acc_code` (a unique identifier), `report_datetime` (unreliable, see Section 2.3), `province_th` (redundant with `province_en`), `route` (a free-text column with too many unique values), and `incident_datetime` (already replaced by its derived features). `number_of_fatalities` and `number_of_injuries` were also excluded from the feature set, since they directly define the target and would otherwise cause data leakage.

3.10 Train-test split

The cleaned dataset was split 80/20 into training and test sets using `train_test_split` with `random_state=42` for reproducibility, and stratified on `accident_severity` so that both sets keep the same class proportions ($\approx 44\%$ PDO, $\approx 44\%$ Injury, $\approx 12\%$ Fatal).

4. Exploratory Data Analysis and Visualization

We used descriptive statistics and visual exploration to understand the temporal, spatial, and contextual patterns present in the cleaned dataset.

4.1 Temporal patterns

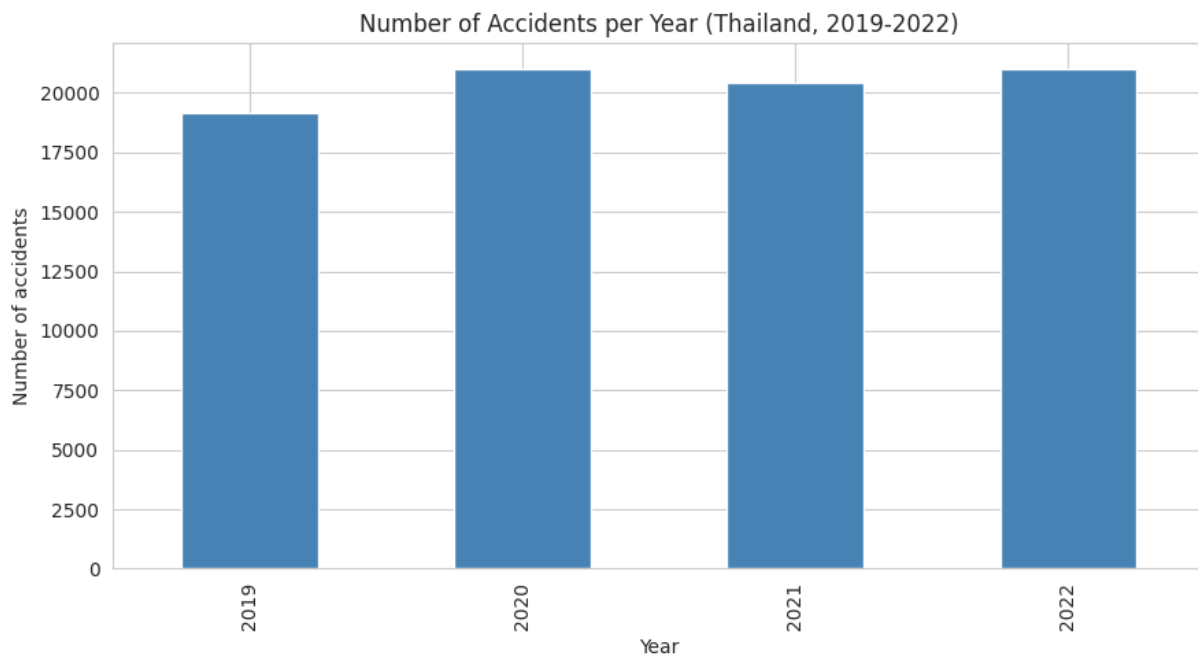


Figure 1. Number of accidents recorded per year (2019–2022).

Accident counts vary across the four years covered. Since reporting coverage may not be perfectly uniform across agencies and years (see Section 9), year-to-year differences should be interpreted as general trends rather than exact accident rates.

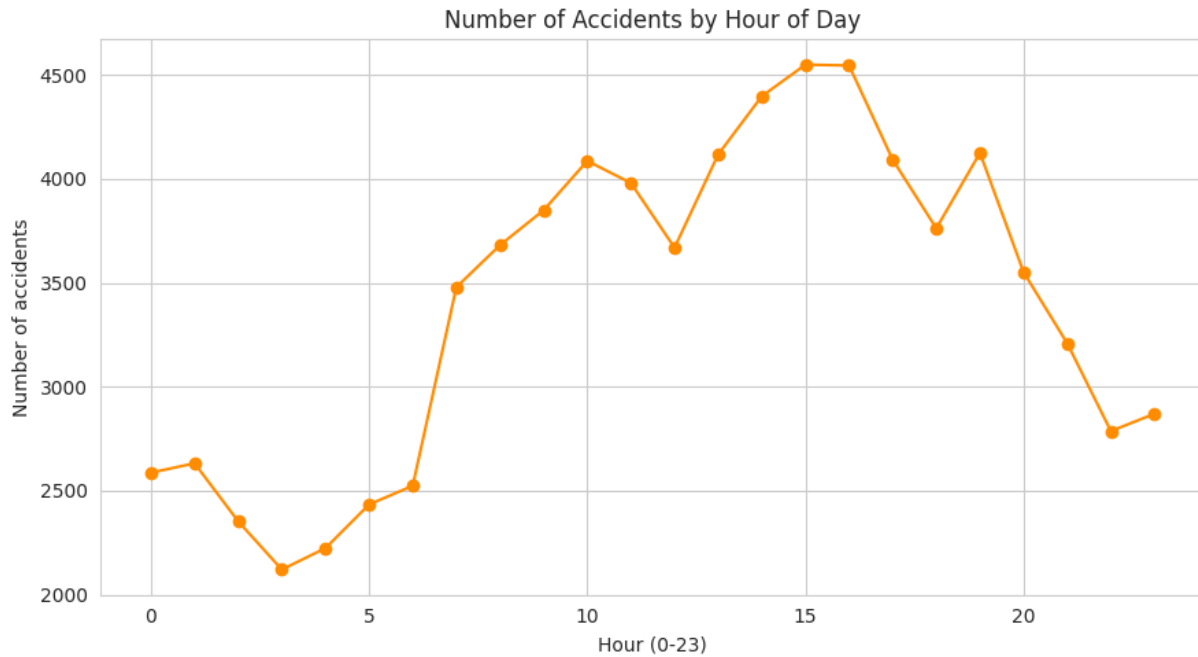


Figure 2. Number of accidents by hour of day.

Accident frequency follows the expected daily traffic pattern, increasing sharply in the morning and again in the late afternoon and evening, which corresponds to commuting hours. Late-night hours have far fewer accidents in total, but as shown later, the accidents that do occur during this period are not necessarily less severe.

4.2 Spatial patterns

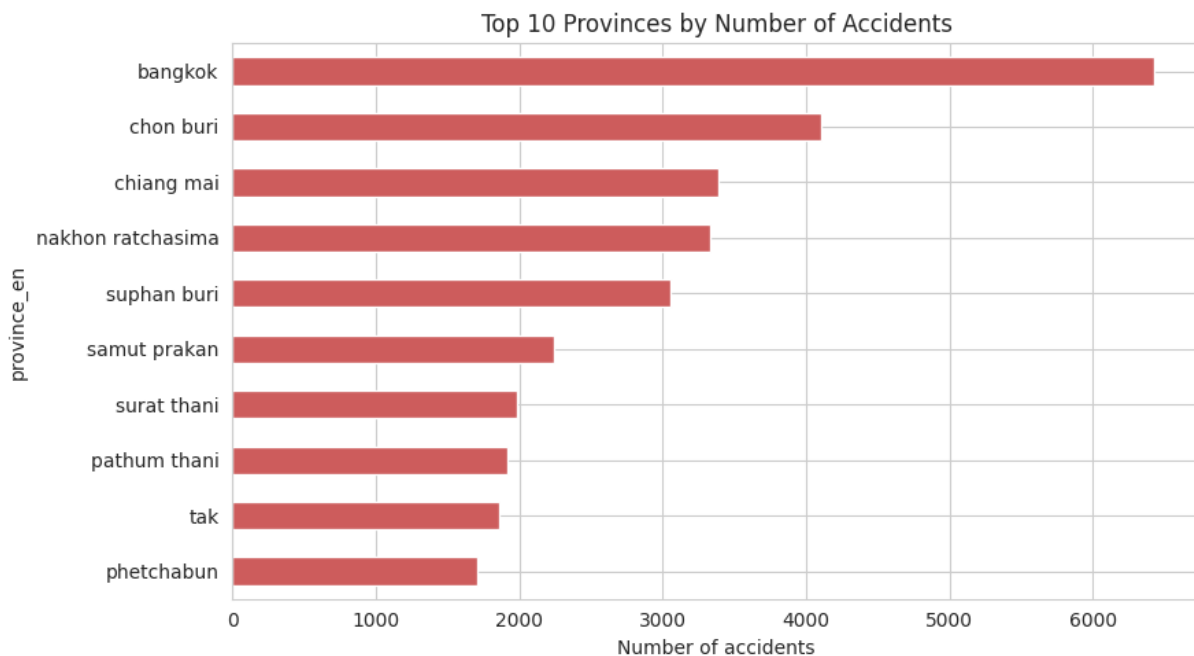


Figure 3. Top 10 provinces by total number of accidents.

Accident counts are concentrated in a small number of provinces, which likely reflects both traffic volume and the length and density of the road network managed by the Department of Highways in these areas.

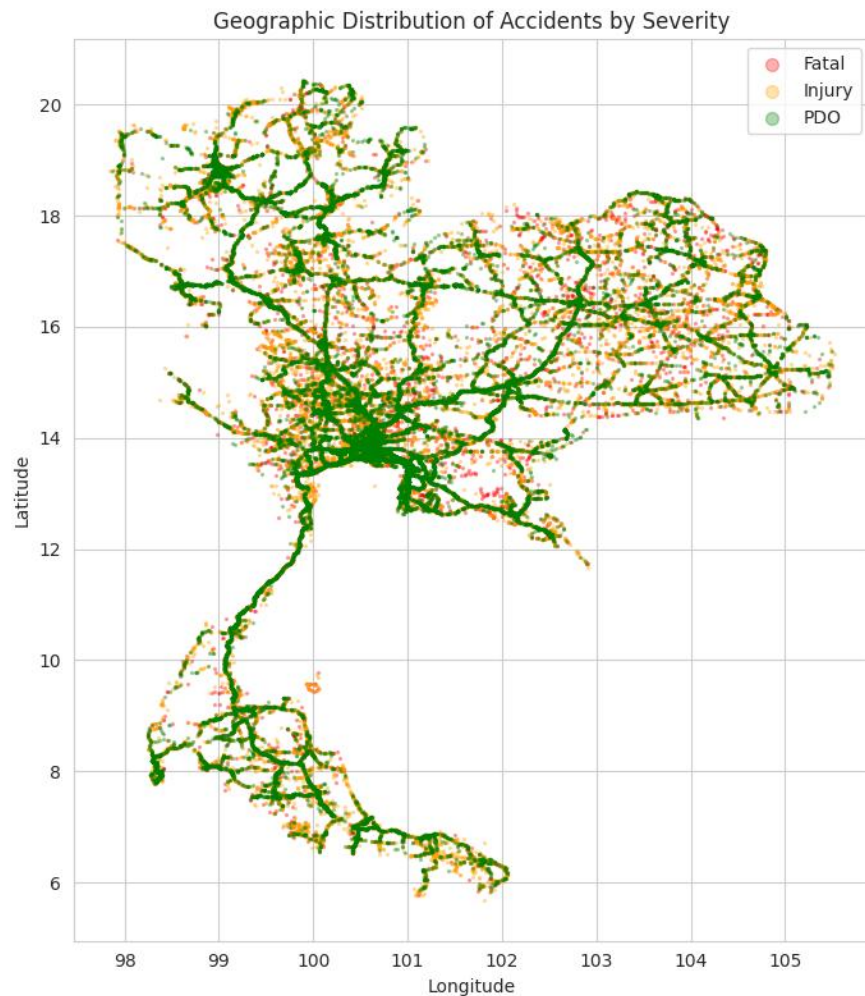


Figure 4. Geographic distribution of accidents by severity class (rows with valid GPS only).

Accidents are concentrated along Thailand's main highway corridors. Fatal accidents (shown in red) are distributed across the whole network rather than concentrated in a single region, which suggests that severity depends more on local conditions, such as speed or road geometry, than on geography alone.

4.3 Severity distribution

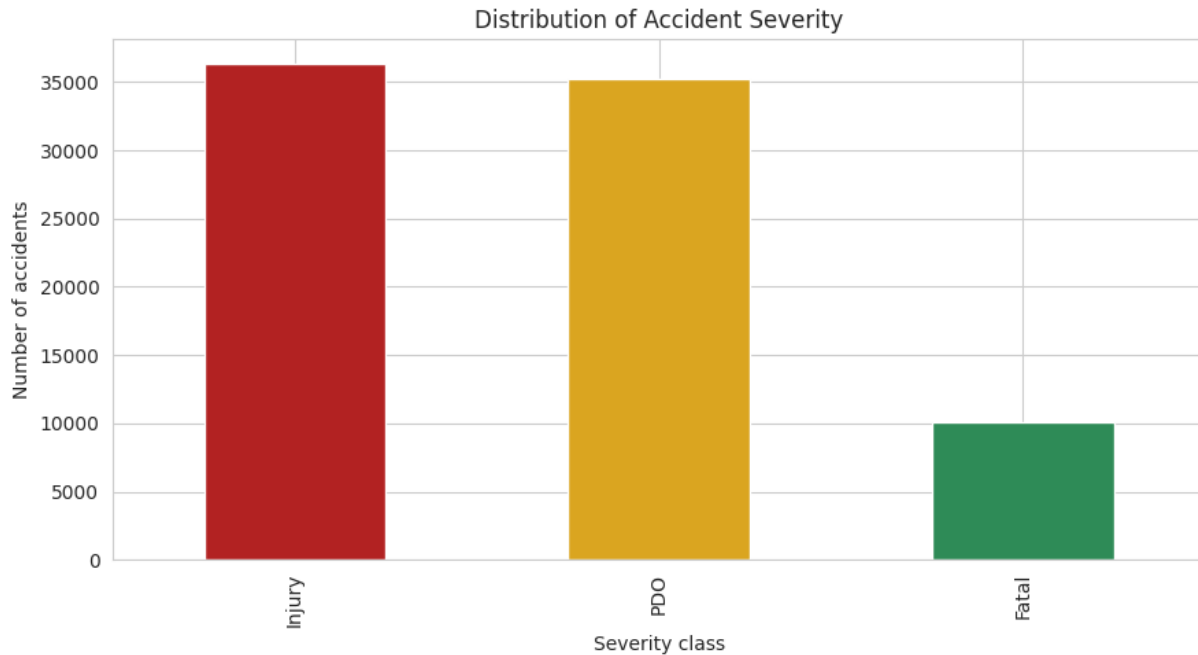


Figure 5. Overall distribution of the engineered accident_severity target.

The three classes are reasonably balanced (PDO 44.4%, Injury 44.5%, Fatal 12.0%), which supports using this variable as a multi-class classification target without needing strong resampling.

4.4 Severity by weather and vehicle type

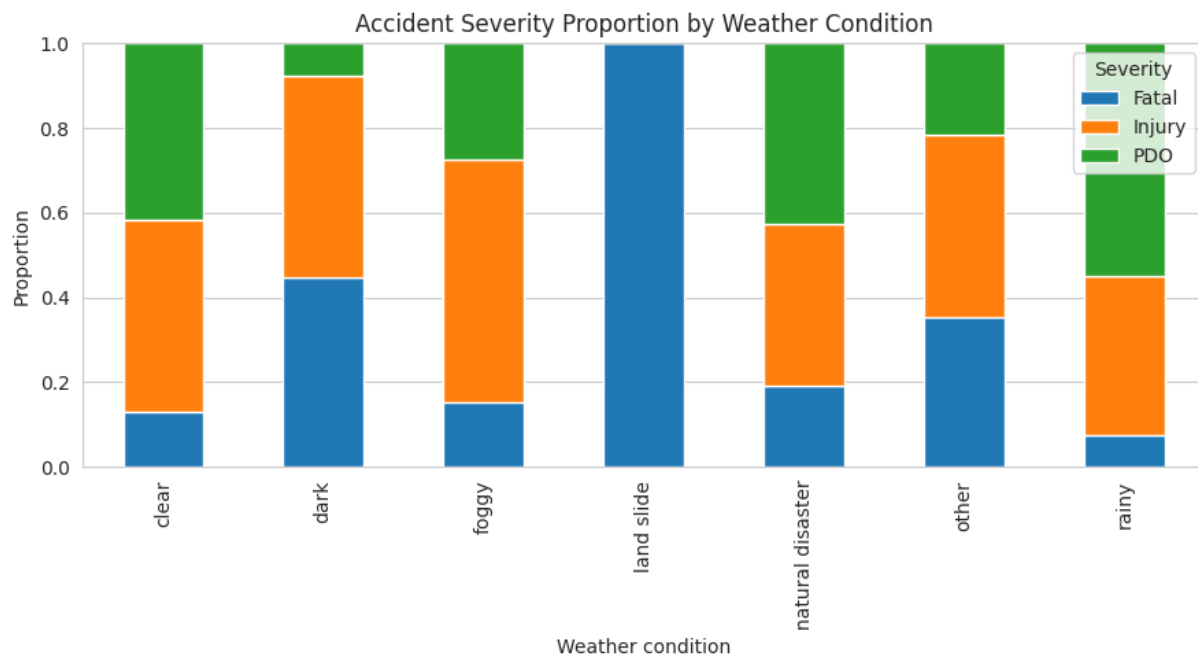


Figure 6. Proportion of severity classes within each weather condition.

Rare weather conditions such as fog and natural disaster events show a visibly higher proportion of fatal outcomes than the dominant “clear” condition, although their small sample sizes mean that this pattern should be considered suggestive rather than conclusive.

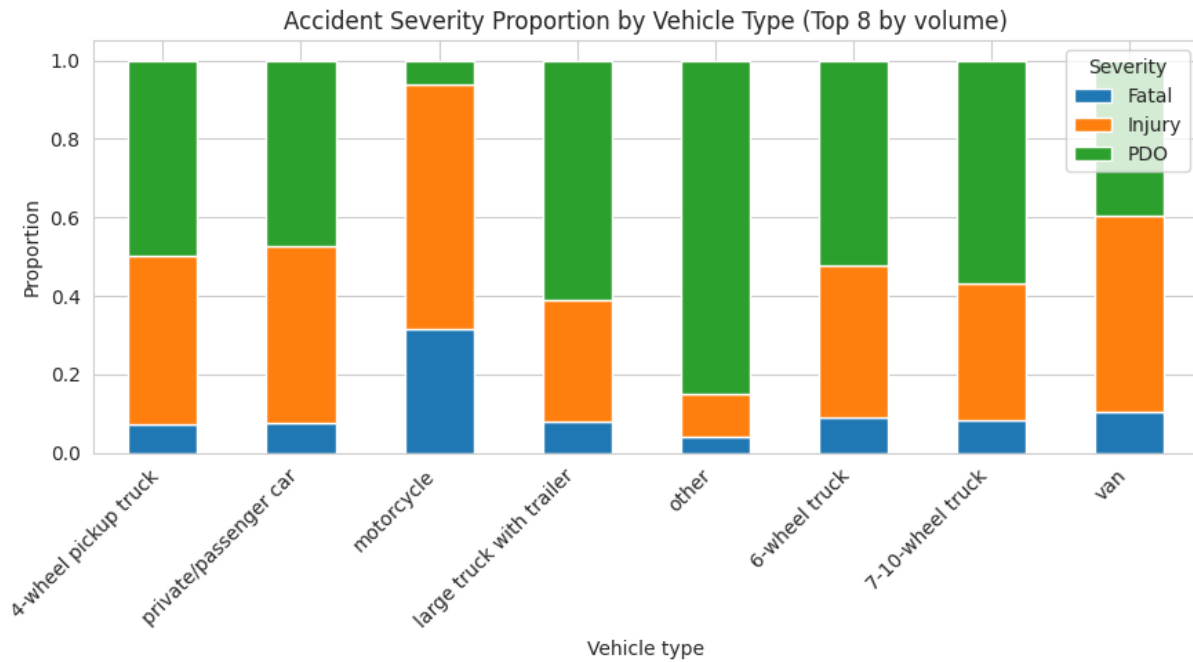


Figure 7. Proportion of severity classes within the top 8 vehicle types by volume.

Vehicle type is clearly associated with severity: motorcycles and large trucks show different severity profiles than passenger cars, which is consistent with differences in occupant protection and crash dynamics.

4.5 Correlation among numeric features

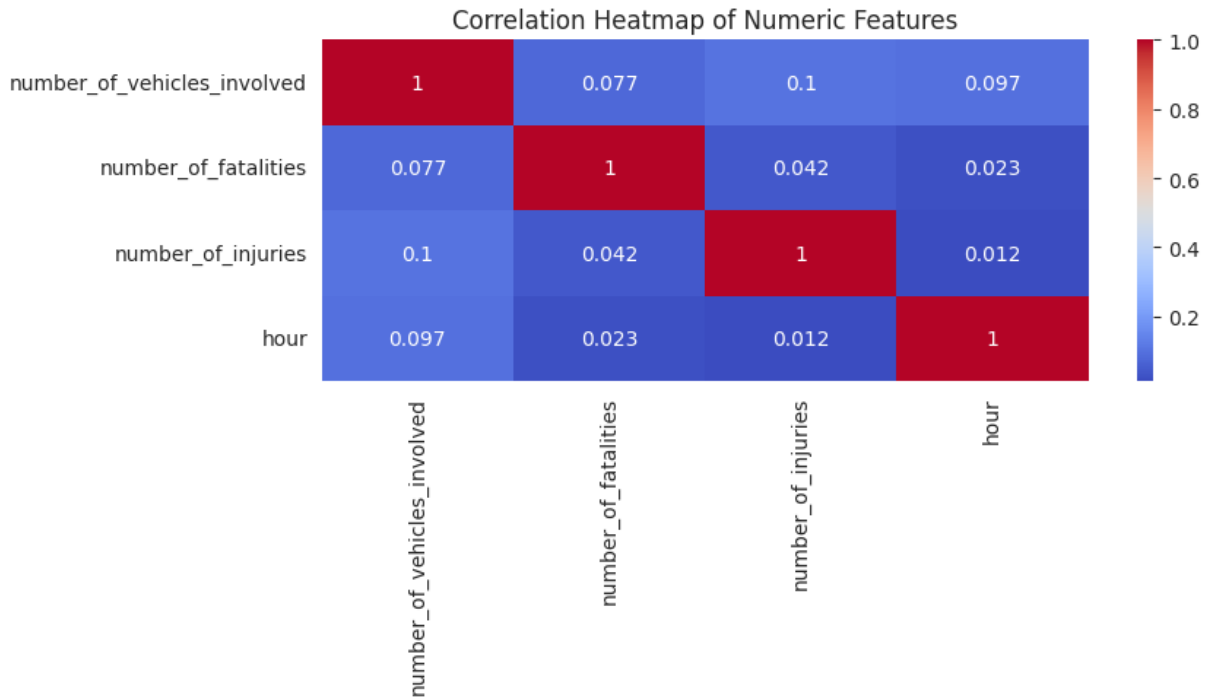


Figure 8. Correlation heatmap of numeric features (vehicles involved, fatalities, injuries, hour).

Fatalities and injuries show only a weak positive correlation with each other and with the number of vehicles involved, which confirms that severity is not simply determined by the number of vehicles involved in the collision. This supports our choice to model severity using a broader set of contextual features rather than relying only on numeric counts.

5. Feature Engineering

Several features were engineered or transformed to prepare the dataset for modeling:

- Time-based features (hour, month, day_of_week, is_weekend) were extracted from the reliable incident_datetime column, to capture daily and weekly traffic and fatigue patterns.
- province_en was reduced to the 15 most frequent provinces, with all other provinces grouped into an “other_province” category, to avoid an excessively sparse one-hot encoding given Thailand’s 77 provinces.
- Categorical variables (vehicle_type, weather_condition, road_description, slope_description, agency, accident_type, day_of_week, province_grouped) were encoded using One-Hot Encoding, since none of these categories have a natural order.
- Numeric variables (number_of_vehicles_involved, hour, month, is_weekend) were standardized with StandardScaler, mainly to benefit Logistic Regression, since tree-based models are not affected by feature scale.
- presumed_cause was deliberately excluded as a feature. Even though it might appear predictive, it represents a post-hoc human judgment about the accident (often defaulting to “speeding”) rather than information available in real time, and using it would risk reinforcing a labeling bias instead of learning genuine risk factors (see Section 9).

All preprocessing steps were implemented as a single scikit-learn ColumnTransformer wrapped in a Pipeline together with each classifier, so that the same fitted transformation (fitted only on the training set) is consistently applied to the test set, which prevents data leakage.

6. Machine Learning Model Training

Three classification models were trained and compared on identically preprocessed data, using a fixed `random_state=42` throughout for reproducibility:

- Logistic Regression (`max_iter=1000`), used as a simple and interpretable linear baseline.
- Decision Tree (`max_depth=10`), which can capture non-linear interactions and threshold effects; the depth was limited to reduce overfitting.
- Random Forest (`n_estimators=200`, `max_depth=15`), an ensemble of decision trees expected to generalize better than a single tree.

Each model was wrapped in the same preprocessing pipeline described in Section 5, to ensure a fair comparison. No hyperparameter besides the depth and estimator counts above was tuned, in order to keep the comparison transparent and reproducible within the scope of this course project.

7. Evaluation Metrics and Results

Since the target variable has three classes with a meaningful minority class (Fatal, 12%), we report accuracy together with macro-averaged precision, recall, and F1-score, which weight all three classes equally instead of letting the majority classes dominate the metric. We also report the full per-class results and confusion matrices, so that the performance on the minority Fatal class cannot be hidden behind a high overall accuracy.

7.1 Overall comparison

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Logistic Regression	0.586	0.568	0.473	0.473
Decision Tree	0.582	0.572	0.490	0.494
Random Forest	0.595	0.604	0.476	0.472

Random Forest achieves the highest overall accuracy and macro-precision, while the Decision Tree achieves the highest macro-recall and macro-F1. The differences between models remain modest (within 1–2 percentage points on most metrics), which suggests that the available contextual features provide a real but limited ceiling on predictive power for this task, regardless of which of these three algorithms is used.

7.2 Per-class performance

Model	Class	Precision	Recall	F1-score	Support
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Logistic Regression	Fatal	0.53	0.11	0.18	2026
Logistic Regression	Injury	0.55	0.62	0.59	7261
Logistic Regression	PDO	0.62	0.69	0.65	7038
Decision Tree	Fatal	0.55	0.19	0.28	2026
Decision Tree	Injury	0.58	0.47	0.52	7261
Decision Tree	PDO	0.59	0.81	0.68	7038
Random Forest	Fatal	0.62	0.09	0.16	2026
Random Forest	Injury	0.57	0.59	0.58	7261
Random Forest	PDO	0.62	0.74	0.68	7038

The per-class results reveal the main honest finding of this project: all three models struggle strongly to recall the Fatal class (recall between 0.09 and 0.19), even though their precision on Fatal is comparatively higher (0.53–0.62). In other words, when a model predicts “Fatal”, it is correct slightly more often than not, but it misses most of the actual fatal accidents, predicting them instead as Injury or PDO. The Decision Tree achieves the best (although still low) Fatal recall at 0.19, which also explains why it leads on macro-F1 despite a lower overall accuracy than Random Forest.

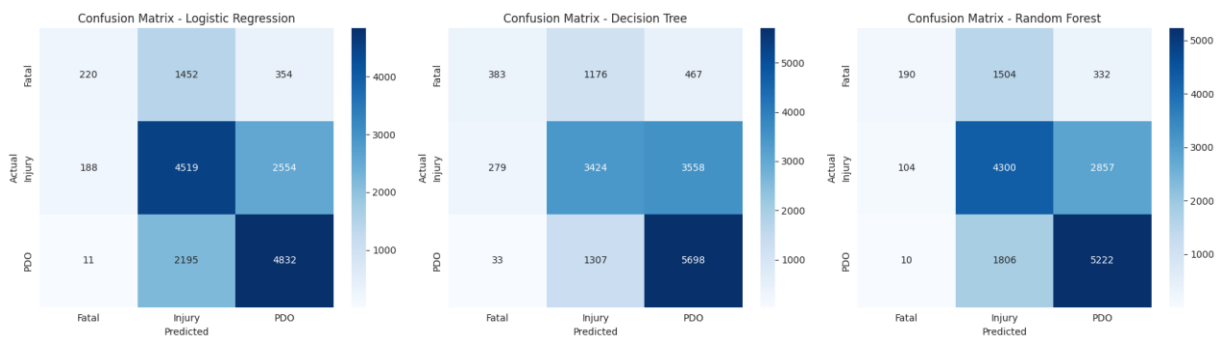


Figure 9. Confusion matrices for Logistic Regression, Decision Tree, and Random Forest (left to right).

The confusion matrices confirm that the Fatal class is most often confused with Injury, which is logical: both involve a person being harmed, and the distinguishing factor, whether the harm was fatal, is not fully captured by the contextual features used here.

7.3 Feature importance

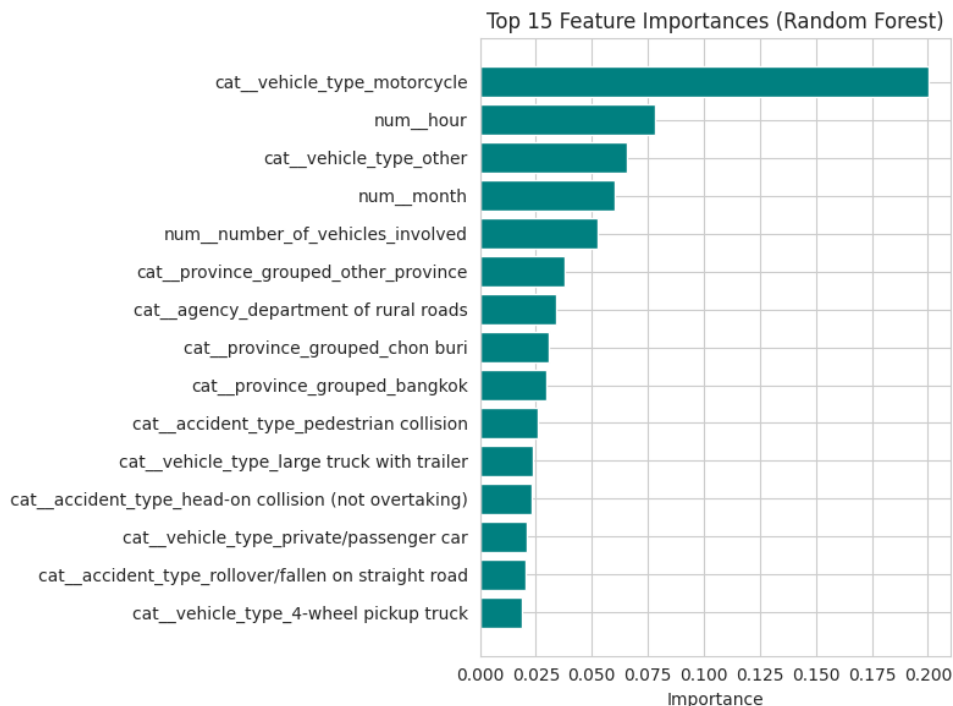


Figure 10. Top 15 feature importances from the Random Forest model.

The number of vehicles involved and several accident-type and vehicle-type categories are among the most influential features, which is consistent with the idea that crash dynamics (head-on versus rear-end, motorcycle versus truck) carry more severity-related information than weather or province alone.

8. Responsible AI Practices

8.1 Data privacy and sensitivity

The dataset does not contain any personally identifiable information: no names, driver IDs, or license plates. Geolocation is recorded only at the accident-site level (latitude/longitude of the event), and is not linked to any individual's identity or movements. We consider the privacy risk of this dataset to be low.

8.2 Bias and fairness concerns

- Cause-labeling bias: presumed_cause is dominated by “speeding” (≈74% of recorded causes), which is more plausibly explained by reporting convenience, meaning it is probably the easiest option for an officer to select, than by a true 74% prevalence of speeding as the actual root cause. We addressed this by excluding presumed_cause from the model entirely, instead of letting the model learn from a potentially biased label.
- Uneven reporting coverage: the three agencies involved (Department of Highways, Department of Rural Roads, Expressway Authority) manage very different volumes of road and may not monitor or report accidents with the same level of consistency. Provinces or years with fewer recorded accidents may therefore reflect weaker monitoring infrastructure rather than genuinely fewer accidents.

- Geographic profiling risk: a severity model that uses province as a feature could be misused to unfairly characterize certain regions or their drivers as inherently more dangerous, when the underlying cause may actually be infrastructure quality or reporting practices rather than driver behavior.

8.3 Data leakage avoidance

`number_of_fatalities` and `number_of_injuries`, the columns used to build the target, were excluded from the feature set. All preprocessing steps (one-hot encoding, scaling) were fitted exclusively on the training split and applied to the test split without re-fitting, using the scikit-learn Pipeline and ColumnTransformer structure, so that no information from the test set influenced preprocessing or model selection.

8.4 Metric choice and class imbalance

Since Fatal accidents represent a minority (12%) but the most serious outcome, we deliberately reported macro-averaged precision, recall, and F1-score, together with full confusion matrices, instead of accuracy alone. Accuracy alone could appear high ($\approx 59\%$) simply by performing well on the majority Injury and PDO classes while almost entirely missing fatal accidents, as shown explicitly in the per-class table in Section 7.2.

8.5 Honest reporting of limitations

- `report_datetime` is unreliable for any time-based analysis (59% of reports were filed more than 30 days after the incident, and one report even precedes the incident itself), and was excluded for this reason.
- All three models show a genuinely weak ability to detect Fatal accidents (recall between 9% and 19%). We do not claim that this model is suitable for real-time severity prediction or emergency triage; it is intended only for descriptive, retrospective, policy-insight purposes.
- The dataset's coverage across the 2019–2022 period and across reporting agencies may not be uniform, which limits the precision of any year-over-year or province-over-province comparison.
- The “other” catch-all category present in `vehicle_type`, `weather_condition`, and `presumed_cause` limits fine-grained analysis of these sub-populations.

8.6 Misuse risk and recommended use

This model should not be used as the sole basis for insurance pricing, individual liability determination, or automated emergency dispatch. It is intended to help identify where road safety efforts should focus, for example which vehicle types or road configurations deserve closer engineering or enforcement attention, not to make automatic decisions about individuals, and always together with the limitations described above.

8.7 Generative AI usage disclosure

Generative AI (Claude, by Anthropic) was used as a learning aid during this project: to help structure the notebook, suggest cleaning and feature-engineering approaches, debug code, and improve the clarity of written explanations. All code, analysis, and conclusions in this report were reviewed and understood by the group members, and no AI-generated text or code was submitted without review.

9. Conclusion and Future Work

9.1 Main findings

- After cleaning, accident severity forms a workable three-class problem (PDO 44.4%, Injury 44.5%, Fatal 12.0%).
- Random Forest achieved the best overall accuracy (0.595) and macro-precision (0.604); the Decision Tree achieved the best macro-recall (0.490) and macro-F1 (0.494); Logistic Regression performed comparably but slightly below both on most metrics.
- Across all three models, the Fatal class remains very difficult to detect (recall 9–19%), which indicates that vehicle type, weather, road geometry, time, and province, although informative, are not sufficient on their own to reliably distinguish fatal from non-fatal outcomes.
- Vehicle type, accident type, and the number of vehicles involved were the most informative features for the Random Forest model.

9.2 Limitations

The most important limitation is the weak recall on the Fatal class, which limits the practical usefulness of this model for prediction. Several relevant factors are simply not present in the dataset, such as impact speed, whether the person wore a helmet or seatbelt, and the time taken for the ambulance to arrive, together with possible labeling bias in the excluded `presumed_cause` field. Reporting coverage may also be uneven across agencies, provinces, and years, as discussed in Section 8.

9.3 Future work

- Incorporate external data such as posted speed limits, traffic volume, or road infrastructure quality ratings to better explain severity.
- Apply class-balancing techniques (exemp, SMOTE or class-weighted loss) specifically to improve recall on the Fatal class, and compare the resulting precision/recall trade-off.
- Build a proper province-level choropleth map and formal spatial hotspot analysis (exemp, Kernel Density Estimation) to complement the scatter plot in Figure 4.
- Investigate whether a binary “Fatal vs. non-Fatal” reframing, rather than three classes, improves recall on the most safety-critical outcome.

10. Team Contribution

The work was divided so that each member contributed code, analysis, written sections, and a meaningful part of the oral presentation:

Member	Code	Report sections	Slides	Oral presentation
Baptiste LORTON	Data quality assessment, data cleaning, target variable engineering, correlation heatmap	2 (Dataset Description), 3 (Preprocessing), 9 (Responsible AI)	3, 4, 5, 14	Introduction, dataset, data quality issues, Responsible AI
Romain BARDET	All EDA visualisations (temporal, spatial, geospatial map, severity ×	4 (EDA & Visualisation), 5	6, 7, 8, 9	Exploratory Data Analysis, Feature Engineering

	weather, severity × vehicle type), feature engineering (time features, province grouping, OneHotEncoder, StandardScaler)	(Feature Engineering)		
Quentin COUPÉ	Train-test split, ML pipeline construction (Logistic Regression, Decision Tree, Random Forest), model training, evaluation metrics, confusion matrices, feature importances	6 (ML Model Training), 7 (Evaluation Metrics & Results)	10, 11, 12, 13	Machine learning model evaluation results, honest limitations

11. References

- Thailand Road Accident (2019–2022) dataset, Kaggle. Originally sourced from the Office of the Permanent Secretary, Ministry of Transport, Thailand.
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